DELAY-EMBEDDING ANALYSIS OF NEURAL NETWORK TRAINING LOGS: PRECONDITIONS, NEGATIVE RESULTS, AND VALIDATED DRIVER RECOVERY

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ABSTRACT

Empirical dynamic modelling promises to recover the structure of a high-dimensional system from scalar observations, which makes it attractive for monitoring neural network training from logs that consume negligible memory. This report establishes when that promise is redeemable. The embedding theorems of Takens and Stark reconstruct a compact invariant set that the trajectory revisits; we measure recurrence directly and find that raw training logs do not satisfy this precondition, with recurrence rates falling to zero under temporal exclusion while a reference chaotic attractor retains 0.93. We then report three candidate signals for the delayed generalisation phenomenon known as grokking, each of which separates training runs convincingly and none of which survives a controlled comparison: an intrinsic-dimension estimate that reproduces closed-form constants of a straight line, a function-space velocity that tracks weight decay rather than generalisation, and a shared-manifold coupling that tracks configuration rather than the transition. Against these we set a positive result obtained in the regime where Stark's theorem does apply. Convergent cross mapping recovers an externally injected driver from a single one-dimensional loss log in seven of eight coupled runs across two architectures, with no false positives in five control runs whose driver is logged but never applied, and with a limitation for strictly periodic drivers that we characterise quantitatively. Finally, a factorial study of thirty runs shows that the delay reported in earlier work lies far outside the distribution produced by its own configuration. We conclude with a control protocol that would have falsified each negative result before publication.

Keywords: empirical dynamic modelling, delay embedding, surrogate data, convergent cross mapping, grokking, training diagnostics.

1 INTRODUCTION

A training run exposes very little of itself cheaply. Weights and activations are large and expensive to store, whereas scalar logs such as the training loss, the validation loss and the parameter norm cost almost nothing. If the geometry of the underlying optimisation could be reconstructed from those scalars, monitoring would become inexpensive and architecture independent. This is the premise of empirical dynamic modelling, and it rests on the embedding theorems of Takens (Takens, 1981) and, for forced systems, of Stark (Stark, 1999; Stark et al., 2003).

Earlier work in this project applied that premise to grokking, the delayed generalisation reported by Power et al. (2022), and proposed that a collapse in the intrinsic dimension of a delay-embedded weight-norm series precedes the transition. The present report re-examines that proposal, reports what replaces it, and states the boundary that separates the two.

The organising claim is the following. The embedding theorems reconstruct a compact invariant set that the trajectory revisits. A training run driven by an external signal with its own attractor satisfies this requirement by construction. The intrinsic optimisation transient does not. Results that respect this boundary reproduce across architectures and controls; results that cross it recover artefacts of trajectory smoothness or of run configuration. Section 2 measures the precondition, section 3 reports three signals that violate it, section 4 reports the positive result obtained inside it, and section 5 re-examines the phenomenon itself. Section 6 states the protocol we recommend.

2 PRECONDITIONS FOR DELAY EMBEDDING

Takens' theorem and its extensions guarantee that a delay map is an embedding of an *invariant set*. The guarantee is asymptotic and geometric: the orbit must return near its past states, so that neighbours in the reconstructed space are dynamically related rather than merely adjacent in time. When this fails, nearest-neighbour statistics measure the local tangent to the trajectory instead of the geometry of an attractor, a distortion first identified by Theiler (1986).

We therefore measure recurrence before invoking any theorem. For a delay-embedded series we fix a neighbourhood radius once, as a low quantile of all pairwise distances, and report the recurrence rate as a function of a temporal exclusion window. Fixing the radius once is essential; recomputing it after each exclusion would force the recurrence rate to equal the quantile for any input whatsoever, an error we made and corrected during this work. The shape of the resulting profile separates the two cases of interest. On an invariant set the rate falls to a positive plateau, because genuine returns occur at all temporal separations beyond the orbit's period. On a transient the rate decays towards zero, because every close pair was adjacent in time.

The estimator is calibrated on systems with known answers. A Lorenz-63 trajectory retains 0.93 of its recurrence rate at the widest exclusion, whereas a monotone ramp, which has no dynamics at all, decays to exactly 0.00. Figure 1 applies it to the training logs.

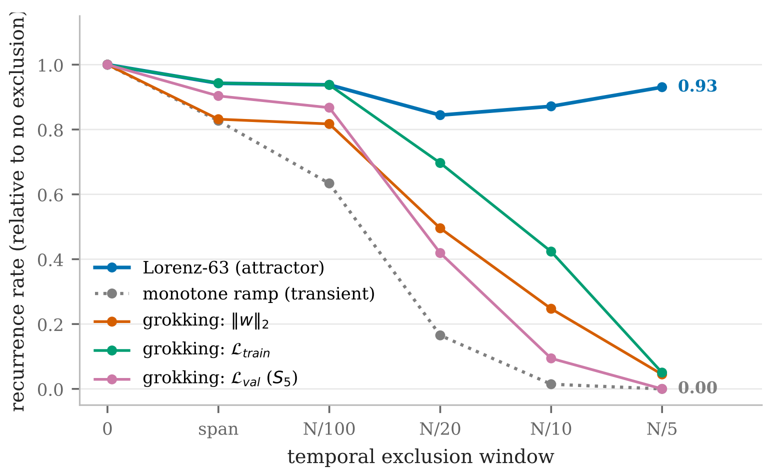

Figure 1: Recurrence rate as a function of the temporal exclusion window, normalised by the rate at zero exclusion. The Lorenz attractor holds a plateau, while all three training observables decay towards the monotone-ramp reference. The training trajectories do not revisit their past states, so no invariant set is available for the embedding theorems to reconstruct.

All three training observables behave like the transient reference rather than like the attractor. The validation loss of the S_5 runs reaches a ratio of 0.000 and the weight norm falls between 0.00 and 0.07. This is a structural fact about optimisation rather than a defect of any particular estimator, and it explains why dimension estimates computed on these logs are not measurements of an attractor.

One caveat constrains the use of this diagnostic. Independent noise also produces close pairs at arbitrary temporal separations and therefore also scores near unity. Recurrence separates smooth transients from everything else; it does not separate an attractor from noise, and a determinism test must follow it. We return to this point in section 3.

3 SIGNALS THAT DO NOT SURVIVE CONTROLLED COMPARISON

3.1 INTRINSIC DIMENSION OF THE WEIGHT-NORM TRAJECTORY

The Levina–Bickel maximum likelihood estimator (Levina & Bickel, 2004) applied to a sliding window of the weight-norm series produces a curve that rises during memorisation and falls before generalisation. Three observations remove its interpretation as a dimension. The first is analytic.

Proposition 1 (The estimate is a constant of a straight line)**.** *Let the delay vectors be sampled at uniform intervals from a curve that is locally straight at the scale of the neighbourhood, and let a Theiler exclusion of W samples be applied. The k nearest neighbours of an interior delay vector are then its temporal neighbours at offsets $\pm(W{+}1), \pm(W{+}2), \ldots$, so the ordered neighbour distances satisfy $r_j \propto s_j$, where s is the non-decreasing sequence $W{+}1, W{+}1, W{+}2, W{+}2, \ldots$ truncated to k terms. The Levina–Bickel estimate*

$$\hat{d} \;=\; \left[\tfrac{1}{k-1}\textstyle\sum_{j=1}^{k-1}\log(r_k/r_j)\right]^{-1} \;=\; (k-1)\Big/\textstyle\sum_{j=1}^{k-1}\log(s_k/s_j) \tag{1}$$

is therefore a function of k and W alone, and contains no information about the data. For $k = 5$ it equals 1.330 *when $W = 0$ and* 10.765 *when $W = 14$.*

The proposition is verified against the estimator in Fig. 2(a). Applied to a synthetic straight line, which possesses no dynamics whatsoever, the estimator returns the predicted constant to within 1.1% at every k and W tested. Applied to the published control runs, it returns the same constants: a median of 1.33 and 1.38 without exclusion against a prediction of 1.33, and 11.30 and 11.69 with exclusion against a prediction of 10.76. The plateaus reported in earlier work are therefore reproduced by a straight line, and cannot be evidence about an attractor.

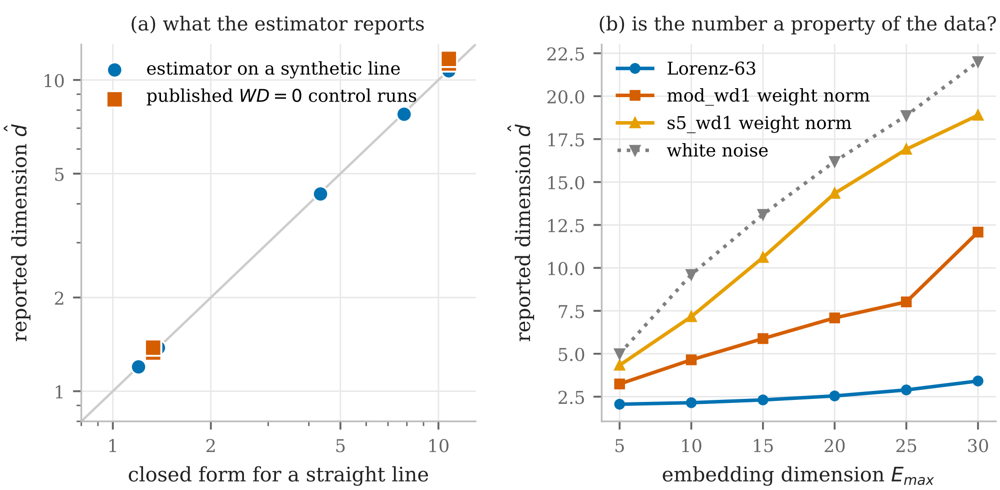

Figure 2: (a) Dimension reported by the estimator against the closed form of Proposition 1, on logarithmic axes with the identity line shown. A synthetic straight line and the published $WD = 0$ control runs both lie on the prediction. (b) Reported dimension against the dimension of the embedding space. A quantity that is a property of the data does not depend on the space it is measured in. The Lorenz attractor rises by a factor of 1.66 across the range, whereas the weight-norm series rise by 3.73 and 4.36, which is the behaviour of the white-noise reference at 4.40.

The second observation concerns identifiability. An intrinsic dimension exists only if the estimate is insensitive to the dimension of the space in which it is measured. Figure 2(b) sweeps that dimension. The Lorenz reference grows by a factor of 1.66 from $E_{max} = 5$ to $E_{max} = 30$, while the two weight-norm series grow by 3.73 and 4.36, which is indistinguishable from the factor of 4.40 obtained for white noise. On this diagnostic the published observable behaves like noise rather than like an attractor, so no dimension is identifiable from it.

The third observation is procedural. The windows were labelled by their centres, so each estimate incorporated data recorded after the step it was assigned to. Under causal labelling, in which an estimate is assigned to the last step it observed, the reported lead over the transition does not survive.

Averaging the per-point estimates also inflates them relative to pooling the likelihood, as MacKay & Ghahramani (2005) note. The correction changes the level but not the conclusion, because the underlying quantity is not identifiable in the first place.

3.2 Function-space velocity

The preceding failure suggests an observable that cannot be a smoothness artefact of the weight norm. We therefore recorded, at each logged step, the logits of a fixed probe set of training inputs, centred each example's logit vector over classes and normalised it to unit length. The resulting series is invariant to the uniform rescaling of logits that weight decay induces, so a change in it indicates that the computed function has moved rather than that its confidence scale has changed. The median per-example movement of the normalised logits separates a grokking run from a run without weight decay by a factor of approximately 360.

The separation does not survive its control. We trained runs in which weight decay is held at 1.0 and only the quantity of training data is reduced, so that the model memorises and never generalises. Figure 3 shows the outcome. The runs that never generalise sustain a *higher* velocity than the runs that grok, and the ordering is consistent across replicate seeds rather than resting on a single draw. The only run far below the others is the one without weight decay, in which the function freezes within a few thousand steps and thereafter only rescales.

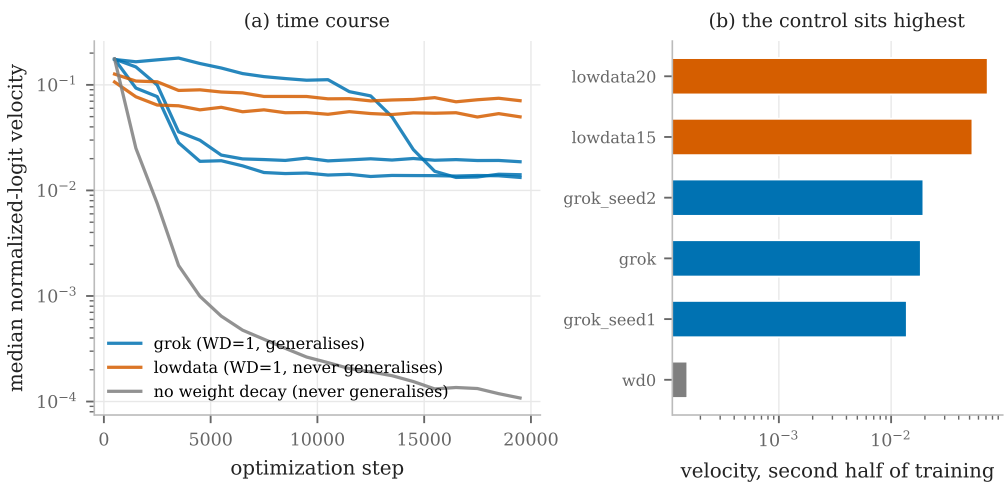

Figure 3: (a) Median velocity of the normalised probe logits against the optimization step, on a logarithmic scale. (b) The same statistic over the second half of training, one bar per run. If the velocity indicated impending generalisation, the runs that never generalise would sit low. They sit highest. Weight decay is identical in the blue and orange families, which differ only in the quantity of training data.

Quantitatively, the runs that never generalise reach 7.26×10^{-2} and 5.30×10^{-2} over the second half of training against 1.38×10^{-2} to 1.94×10^{-2} for the three grokking runs, while the run without weight decay falls to 1.53×10^{-4}. Every run with weight decay lies within a factor of five of every other, and the factor of 360 quoted above is entirely accounted for by the presence or absence of regularisation. The statistic therefore measures the reorganisation that weight decay imposes on the function, and among regularised runs it is, if anything, inversely related to generalisation.

3.3 Shared-manifold coupling

Both preceding failures are statistics of a single transient series. Cross mapping asks a different question, namely whether two observables lie on a common manifold, and that question is well posed on a transient because it requires no recurrence of either series alone. We therefore tested whether the training loss and the parameter norm, both available without any validation data, share a manifold, and whether the coupling changes at the transition.

The result appeared decisive. Figure 4(a) shows a clean separation with no overlap: the three runs with a delayed transition have median coupling between 1.00 and 1.15, while the four runs without lie between 2.87 and 5.61.

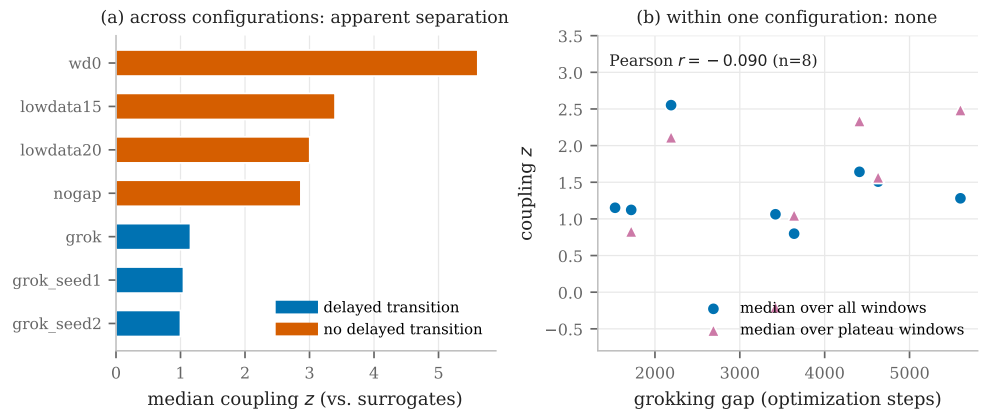

Figure 4: (a) Median coupling between the training loss and the parameter norm separates runs with a delayed transition from runs without. (b) The same statistic across eight runs of a single configuration, in which only the split and initialisation seeds vary. The gap spans 1290 to 5600 steps and the coupling is unrelated to it. The separation in panel (a) reflects configuration, not the transition.

Two checks dissolve it. Within individual runs the statistic does not move at the moment of generalisation; the apparent step in an aggregate over runs arises from averaging runs whose transitions occur at different steps. More importantly, the grouping in panel (a) coincides exactly with configuration, since the low group consists of the runs with a training fraction of 0.3 and weight decay of 1.0 and the high group contains every other setting. We therefore held configuration fixed and varied only the split and initialisation seeds, which the study of section 5 shows produce gaps between 1290 and 5600 steps. The expected sign was recorded before the runs were executed. The observed correlation between the gap and the coupling is -0.090 (Pearson) and $+0.214$ (Spearman) over eight runs, that is, nothing, and for plateau windows it is weakly positive, which is the opposite of the prediction. Within one configuration the coupling spans 0.80 to 2.55, against 1.0 to 5.6 across configurations.

3.4 A COMMON FAILURE MODE

The three signals fail in the same way, as table 1 summarises. Each separates *configurations* rather than *transitions*, and each is supported by a comparison in which the runs differ in more than the property of interest. The controls that falsify them share every feature with the positive runs except the mechanism under test: weight decay held fixed while data is varied, or configuration held fixed while seeds are varied. We regard this, rather than any individual negative result, as the principal methodological lesson.

Table 1: The three signals, the comparison that made each convincing, and the control that falsifies it. In every case the falsifying control differs from the positive runs in one respect only.

Signal	Apparent evidence	Falsifying control	What it measures
Intrinsic dimension of $\|w\|_2$	Rises then collapses before generalisation	A straight line reproduces both plateaus; the estimate tracks E_{max} as white noise does	Local smoothness of the series
Function-space velocity	Separates $WD = 1$ from $WD = 0$ by a factor of 360	$WD = 1$ with reduced data never generalises, yet sustains a higher velocity	Reorganisation imposed by weight decay
Shared-manifold coupling	Separates runs with a delayed transition from runs without, with no overlap	Across eight runs of one configuration the coupling is unrelated to the delay, $r = -0.090$	Run configuration

4 Driver recovery in the regime where the theorems apply

Stark's extension of Takens' theorem covers a system driven by an external signal that possesses its own attractor (Stark, 1999; Stark et al., 2003). A training run modulated by an injected driver satisfies that hypothesis by construction, which makes it the natural setting in which to ask whether anything at all is recoverable from a one-dimensional log.

4.1 Design

We use two independent settings. The first consists of existing logs from a ResNet-18 trained on CIFAR-10, in which a fraction of each batch's labels is corrupted according to a driver logged alongside the loss. The second reproduces the design in this work, using a one-layer transformer on modular addition with drivers under our control. In both cases the ground truth is known, and in both cases the essential control is a *ghost* driver, which is logged exactly like a real one while the batch is returned unmodified. A ghost run shares the architecture, the schedule, the loss trajectory and the logging cadence of a coupled run and differs only in whether coupling exists.

Detection uses convergent cross mapping (Sugihara et al., 2012): the response is delay embedded and used to predict the driver, on the reasoning that if the driver forces the response then the reconstructed manifold of the response carries information about the driver's state. Significance is assessed against two nulls. The first uses iterative amplitude-adjusted Fourier surrogates (Theiler et al., 1992; Schreiber & Schmitz, 1996), which preserve the power spectrum and the exact value distribution of the driver while destroying its alignment with the response. The second rotates the driver in time, which preserves every value and changes only the alignment. No result is reported without them.

4.2 Results

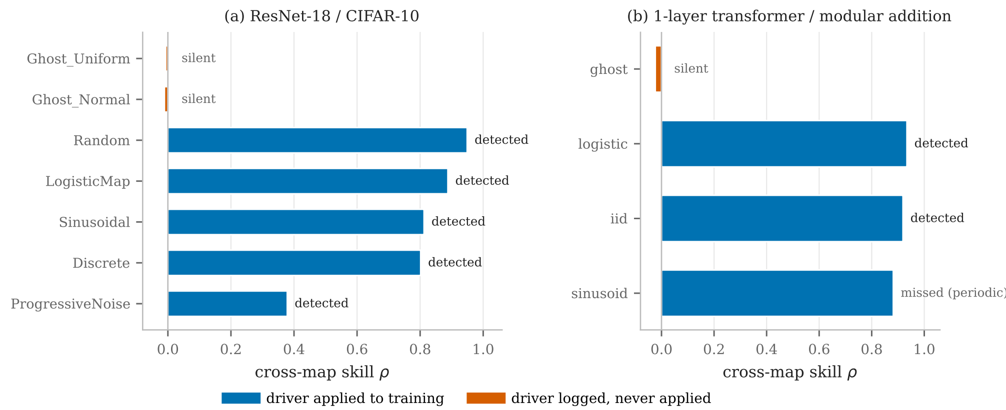

Figure 5: Cross-map skill for every run in both settings. Drivers that were applied to training are recovered from the loss log alone; drivers that were logged but never applied are not. The single missed detection is the smooth periodic driver discussed in section 4.3.

Figure 5 summarises both settings. In the ResNet setting all five coupled drivers are detected, with skills between 0.379 and 0.949 and $p \leq 0.05$ under both nulls, and all ghost runs remain silent at skills of -0.011 and -0.007. In the transformer setting the chaotic and independent drivers are detected at 0.934 and 0.919 and the ghost remains silent at -0.024. Across both settings, seven of eight coupled runs are detected and none of the five ghost runs produces a false positive.

The ghost runs carry the argument. They share the training trend with the coupled runs, so their near-zero skill excludes the explanation that defeated the signals of section 3, namely that a statistic separates runs because their loss curves have different shapes.

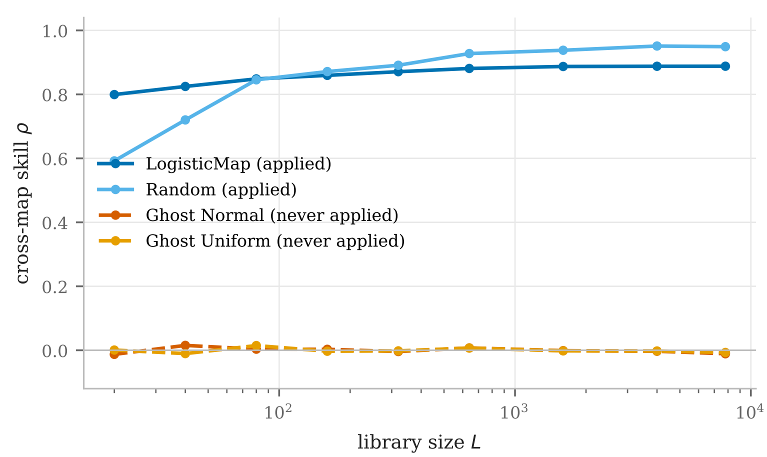

Figure 6: Cross-map skill against library size. Coupled runs improve as the manifold is sampled more densely, which is the convergence signature that distinguishes causation from coincidental correlation. Ghost runs remain at zero for every library size.

Figure 6 shows the convergence criterion that gives the method its name. The skill of the coupled runs rises with library size, most clearly for the independent driver, which climbs from 0.59 to 0.95. The ghost runs show no such rise, and no level.

One preprocessing choice deserves emphasis because it initially inverted the conclusion. Differencing the response is the obvious way to remove the training trend, but it is a high-pass filter and therefore removes slow drivers. The periodic drivers have the strongest direct coupling of any run, with correlations of 0.69 and 0.58 against the loss, yet they cross map at 0.09 and 0.12 after differencing and at 0.81 and 0.80 without it. The trend requires no manual removal, because both nulls and the ghost runs already control for it.

4.3 A LIMITATION FOR PERIODIC DRIVERS

The one missed detection is instructive and is not repaired by a better statistic. The smooth sinusoidal driver of the transformer setting yields a cross-map skill of 0.882, which is high, but its surrogates reach between 0.87 and 0.94, which is equally high. The reason is structural. The loss reveals the phase of the cycle, and any series of the same period, whether a spectrum-preserving surrogate or the driver rotated in time, is a deterministic function of that phase and is therefore equally predictable from it. Spectrum-preserving nulls consequently have no power against a strictly periodic driver. The corresponding driver in the ResNet setting is detected only because it is quantised to ninety-five distinct values, so its surrogates depart from it considerably further. A different null, for example a driver of a different period, would be required.

4.4 THE METHOD IS NOT DISCRIMINATIVE BETWEEN A RUN'S OWN METRICS

For completeness we applied the same procedure to pairs of observables within single uninjected runs. Eleven of sixteen directed pairs pass both nulls, with skills as high as 1.000. This is consistent with the theory, since the training loss, the validation loss and the parameter norm are three observations of one system, but it means that cross mapping between a run's own metrics carries no information about the transition and should not be interpreted as a discovery.

5 THE DELAYED TRANSITION IS NOT TYPICAL OF ITS CONFIGURATION

The configuration used in earlier work seeds a single random stream from which the train and validation split is drawn and from which the model initialisation then continues. Consequently a change of seed alters which examples are used for training and what the initial weights are, and the two cannot be distinguished. We separated them by restarting the stream between the two steps, verified that the modification is inert when unused and that it decouples the two factors in both directions, and then ran a factorial study of six splits by five initialisations, with weight decay fixed throughout.

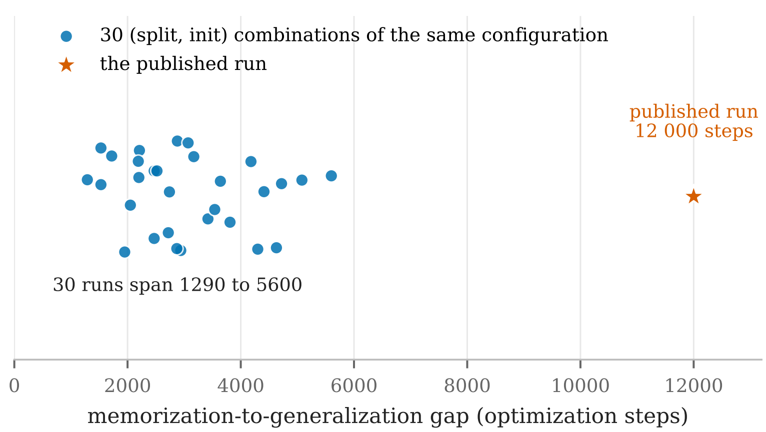

Figure 7: Memorisation-to-generalisation gap for thirty runs of one configuration, varying only the split and initialisation seeds. Every run generalises and every gap lies between 1290 and 5600 steps. The delay reported in earlier work lies far outside this distribution.

All thirty runs generalise, with gaps between 1290 and 5600 steps, a median of 2875 and a standard deviation of 1130. Figure 7 places the previously reported delay of approximately 12 000 steps against that distribution. Five of the thirty runs use the same split as the published run, so the discrepancy is not a property of the split alone.

The variance decomposition is inconclusive by design rather than by accident. The split accounts for 19.8% of the variance in the gap and the initialisation for 23.8%, with the remainder in the interaction. Neither factor dominates. We note that a partial analysis of the first fifteen cells indicated the opposite, with the split accounting for 43% against 17% for the initialisation, which is a caution against reading factorial studies before they complete.

6 Recommended protocol

The following four requirements would have falsified each negative result of section 3 before it was written up, at a cost of well under an hour of computation each.

1. **State and measure the precondition.** Report recurrence before invoking an embedding theorem. If the trajectory does not revisit its past states, the language of attractors and their dimensions does not apply, whatever number an estimator returns.
2. **Test every claim against surrogates.** A statistic that does not exceed an amplitude-adjusted Fourier surrogate ensemble is reporting the linear autocorrelation of the series. Where the driver is periodic, add a rotation null, and note that neither has power in the strictly periodic case.
3. **Include a control that shares everything but the mechanism.** The ghost drivers of section 4 are the model: identical architecture, schedule, loss trajectory and logging, differing only in whether coupling exists. Controls that differ in several respects at once cannot discriminate, which is precisely how the signals of section 3 appeared convincing.
4. **Hold configuration fixed and vary only seeds, with the expected sign recorded in advance.** Any statistic proposed as a predictor of the transition must correlate with the delay across runs of one configuration. This test is inexpensive and it falsified the most promising of our candidates.

Two further practices proved necessary in the course of this work. Estimators should be calibrated against systems with known answers before being applied to project data, since two of our own diagnostics were found to be circular or mis-specified in exactly that way. Preprocessing should be justified per series rather than adopted globally, since subtracting a moving average manufactures structure in sparse series and differencing removes slow drivers.

7 Conclusion

Delay-embedding methods are applicable to neural network training logs in a narrower regime than the earlier proposal assumed, and within that regime they work well. When an external driver possessing its own attractor forces training, convergent cross mapping recovers it from a single scalar loss log, in seven of eight coupled runs across two architectures and with no false positives among five runs whose driver was logged but never applied. The recovered quantity is causal rather than correlational, it requires no validation data, and it is obtained from a log that costs nothing to store.

The intrinsic optimisation transient does not admit the same treatment. It does not revisit its past states, and the three statistics we examined that appeared to detect the grokking transition were found instead to separate run configurations. The phenomenon itself is also less stable than previously supposed, since thirty runs of the published configuration produce delays between 1290 and 5600 steps against a reported value of approximately $12\,000$.

We therefore recommend that future work on this problem either restricts itself to the driven regime, where the theory applies and validation is possible, or targets the question that the present negative results leave open, namely whether any training-side observable distinguishes a run that will generalise from a run that will not when configuration is held fixed. None of the observables examined here, comprising the parameter norm, the trajectory roughness, the intrinsic dimension, the function-space velocity and the shared-manifold coupling, is able to do so.

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